

The nn-bounded operators are not closed under nn-convergence

A counterexample to a question in arXiv:math/0004049

Source Question

Troitsky defines nn-convergence for operators on a topological vector space and asks whether the class of nn-bounded operators is closed relative to that convergence [1].

The class of all nn-bounded operators will be usually equipped with the topology of **nn-convergence**, defined as follows. We will call a collection $\mathcal{G}$ of operators **uniformly nn-bounded** if there exists a base $\mathcal{N}_0$ of zero neighborhoods such that for every $U \in \mathcal{N}_0$ there exists a positive real β such that $S(U) \subseteq \beta U$ for each $S \in \mathcal{G}$. We say that a sequence (S_n) **nn-converges** to zero if there is a base $\mathcal{N}_0$ of zero neighborhoods such that for every $U \in \mathcal{N}_0$ and every $\varepsilon > 0$ we have $S_n(U) \subseteq \varepsilon U$ for all sufficiently large n .

Question. *Is the class of all nn-bounded operators closed relative to nn-convergence?*

Figure 1: The definition of nn-convergence and the displayed question.

The source paper also records that the left shift on $\mathbb{R}^{\mathbb{N}}$, with the topology of coordinatewise convergence, is continuous but not nn-bounded.

Example 2.5. *A continuous but not nn-bounded operator.* Let T be the left shift on the space of all real sequences $\mathbb{R}^{\mathbb{N}}$ with the topology of coordinate-wise convergence, i.e., $T: (x_0, x_1, x_2, \dots) \mapsto (x_1, x_2, x_3, \dots)$. Clearly T is continuous. We will show that T is not nn-bounded. Assume that for every zero neighborhood U in some base $\mathcal{N}_0$ there is a positive scalar α such that $T(U) \subseteq \alpha U$. Since the set $\{x = (x_k) : |x_0| < 1\}$ is a zero neighborhood, there must be a base neighborhood $U \in \mathcal{N}_0$ such that $U \subseteq \{x : |x_0| < 1\}$. Since $T(U) \subseteq \alpha U$ for some positive α then $T^n(U) \subseteq \alpha^n U$, so that if $x = (x_k) \in U$ then $T^n x \in \alpha^n U$, so that $|x_n| = |(T^n x)_0| < \alpha^n$. Hence $U \subseteq \{x : |x_n| < \alpha^n \text{ for each } n > 0\}$. But this set is bounded, while the space is not locally bounded, a contradiction.

Figure 2: Example 2.5 in the source paper.

Proof Intuition

The topology of $\mathbb{R}^{\mathbb{N}}$ only tests finitely many coordinates at a time. Finite truncations of the left shift are harmless: each one has finite-dimensional range, hence maps a suitable zero neighborhood into a bounded set. But for any fixed finite coordinate test, a far enough truncation agrees with the full left shift on all tested coordinates. Thus the truncated shifts converge to the full left shift in nn-convergence even though the full shift is not nn-bounded.

Counterexample

Theorem 1. *The class of nn-bounded operators need not be closed relative to nn-convergence.*

Proof. Let $X = \mathbb{R}^{\mathbb{N}}$ with the product topology, where coordinates are indexed by $0, 1, 2, \dots$. A basic zero neighborhood is a set

$$U(F, \delta) = \{x \in X : |x_j| < \delta_j \text{ for every } j \in F\},$$

where $F \subset \mathbb{N}$ is finite and each $\delta_j > 0$. A subset of X is bounded if and only if it is bounded in each coordinate.

Let $L : X \rightarrow X$ be the left shift

$$L(x_0, x_1, x_2, \dots) = (x_1, x_2, x_3, \dots).$$

For $N \geq 0$, let P_N be the coordinate projection

$$P_N(y_0, y_1, \dots) = (y_0, \dots, y_N, 0, 0, \dots),$$

and put $S_N = P_N L$. We first show that each S_N is nn-bounded. In fact it is nb-bounded. The zero neighborhood

$$V_N = \{x \in X : |x_j| < 1 \text{ for } 1 \leq j \leq N+1\}$$

is mapped by S_N into the set of all y satisfying $|y_j| < 1$ for $0 \leq j \leq N$ and $y_j = 0$ for $j > N$. This image is coordinatewise bounded, hence bounded in X . Thus S_N is nb-bounded, and by the hierarchy in [1, Proposition 2.1], each S_N is nn-bounded.

Now we prove that $S_N \rightarrow L$ in nn-convergence. It is enough to use the standard base $U(F, \delta)$ above. Fix such a $U = U(F, \delta)$ and fix $\varepsilon > 0$. Choose $N \geq \max F$, with the convention that the condition is empty if $F = \emptyset$. For $j \in F$, the j -th coordinate of $(L - S_N)x$ is zero for every $x \in X$, since $P_N L$ keeps all coordinates $0, \dots, N$ of Lx . Therefore

$$(L - S_N)(U) \subseteq \varepsilon U$$

for all sufficiently large N . Hence $S_N - L$ nn-converges to zero, or equivalently S_N nn-converges to L .

Finally, L is not nn-bounded. For completeness we recall the short argument. Suppose L were nn-bounded. Then there would be a base $\mathcal{N}_0$ of zero neighborhoods such that for every $U \in \mathcal{N}_0$ one has $L(U) \subseteq \alpha U$ for some $\alpha > 0$. Choose $U \in \mathcal{N}_0$ with

$$U \subseteq \{x \in X : |x_0| < 1\}.$$

Then $L^n(U) \subseteq \alpha^n U$, and for every $x \in U$ we get

$$|x_n| = |(L^n x)_0| < \alpha^n \quad (n \geq 1).$$

Thus U is contained in a coordinatewise bounded set, so U itself is bounded. But $\mathbb{R}^{\mathbb{N}}$ with the product topology has no bounded zero neighborhood: every zero neighborhood leaves some coordinate unrestricted. This contradiction shows that L is not nn-bounded.

The sequence (S_N) therefore consists of nn-bounded operators, nn-converges to L , and has a non-nn-bounded limit. $\square$

Verification and Scope

No computational verification is needed. The only structural facts used are the coordinatewise description of bounded sets in the product topology and the nb-bounded implies nn-bounded implication from the source paper. The result answers the displayed closure question negatively.

Novelty Check

The run indexes were searched for 0004049, `Spectral Radii`, `nn-bounded`, `nn-convergence`, `left shift`, `product topology`, and `Troitsky`; no prior local packet or attempt was found. Web searches for the exact displayed question and close variants involving `nn-bounded`, `nn-convergence`, and `left shift` did not reveal a separate known answer. This is a bounded novelty check, not an exhaustive literature survey.

References

- [1] Vladimir G. Troitsky, *Spectral Radii of Bounded Operators on Topological Vector Spaces*, arXiv:math/0004049, 2000. <https://arxiv.org/abs/math/0004049>